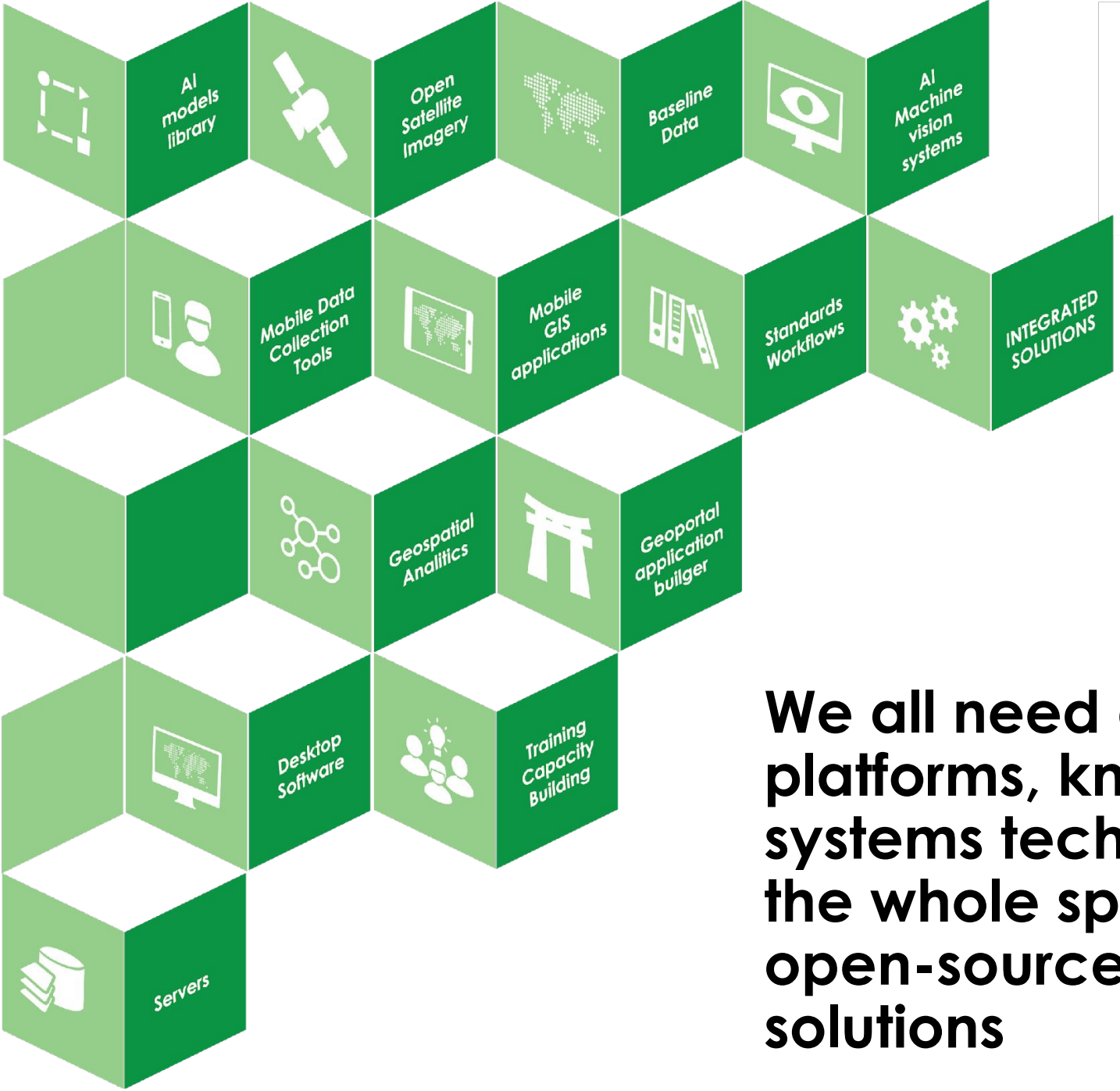
Hybrid GIS Architecture

United Nations Geospatial Information Section
TOS/OSD/OICT

UN Open GIS
INITIATIVE

We all focus on different needs and issues, but we need the same for easy access to open-source technology



UN Open GIS
INITIATIVE

We all need access to platforms, knowhow, systems technology, the whole spectrum of open-source GIS solutions

UN Open GIS Initiative provides **a unique opportunity** to address **Geospatial Technology requirements** to wider **UN Community** and, through the United Nations, **to its operational partners** (NGOs and humanitarian community) but also **to the developing countries** and **academia.**

UN requirements for Hybrid GIS Architecture

GIS enables the Situational Awareness Platforms

- Geospatial visualisation for improving situational awareness
- Geospatial analysis for supporting decision making
- Smart knowledge-based platform for mission operations

GIS supports the fulfilment of core mandates

- Monitors ceasefire agreement & armed groups activities
- Protection of civilians, electoral assistance, etc.

GIS saves lives and supports emergency operations

- Identify locations, and search & rescue operations
- Crisis management, Evacuations
- Go-NoGo, and minefield area

GIS enables cost-effective operations

- Minimise ground visits at planning and operational stages
- Better understanding of mission's operating environment
- Ground water exploration, etc.

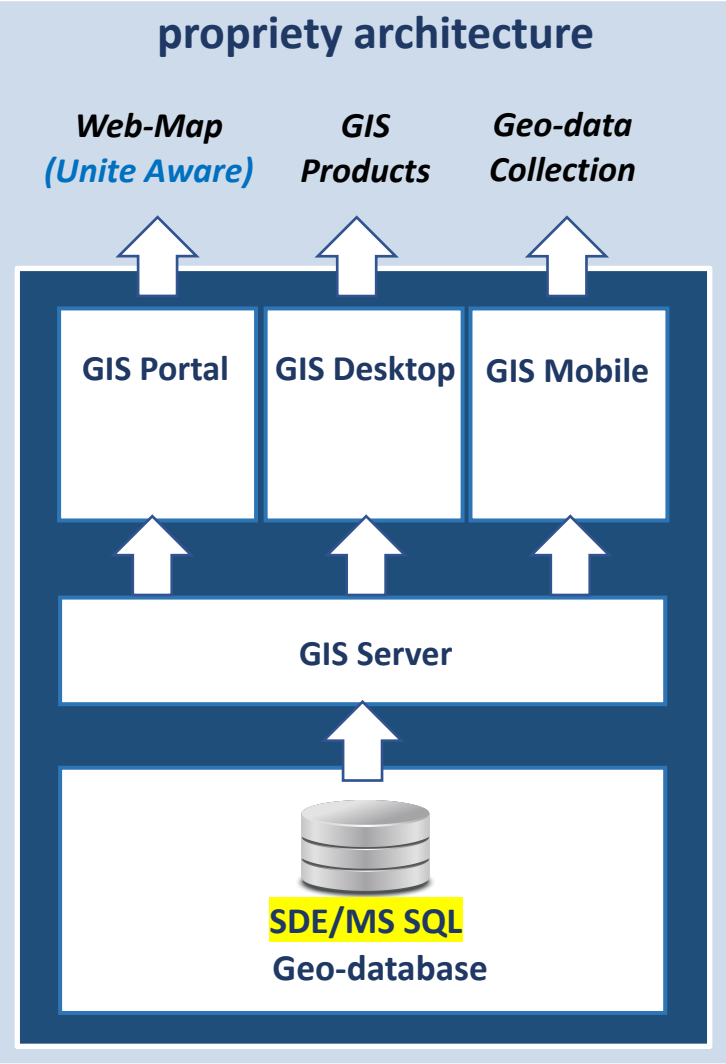
Legacy Systems and Challenges

CURRENT SITUATION:

- UN GIS infrastructure has been built on perpetual software
- **Open-source** used on **add-hoc** bases (Unite Aware OpenStreetMap, QGIS and mobile)
- GIS clients require both **propritery** and **open-source**

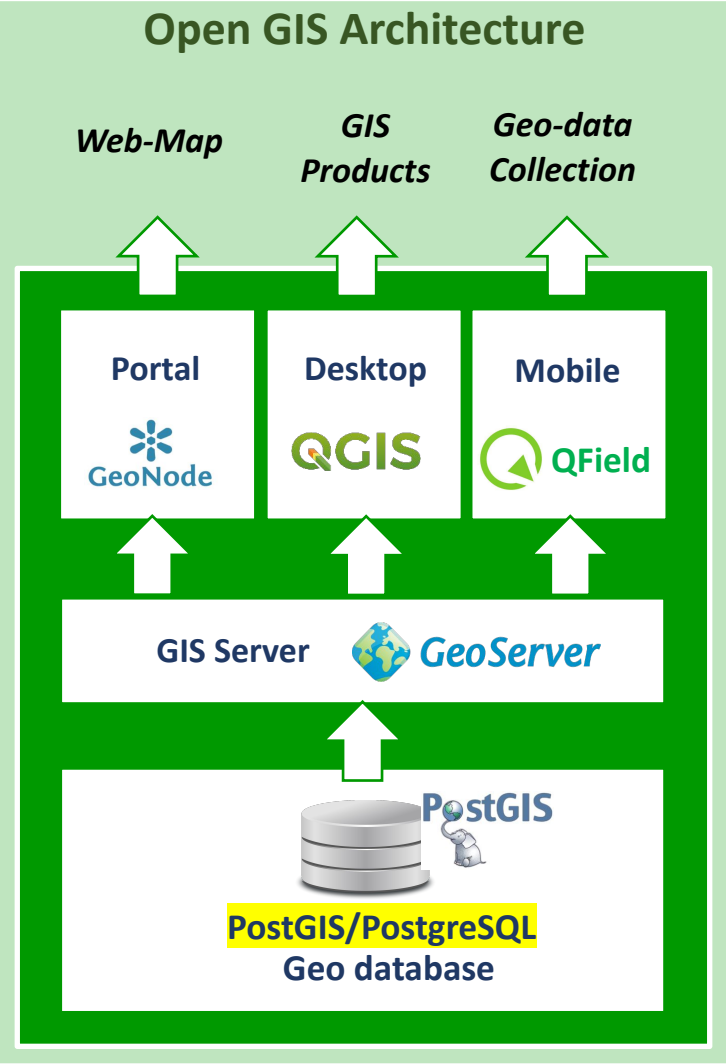
CHALLENGES:

- **Limited flexibility**
- **Restrictions in data formats**
- **High cost of licenses**
- **Limited options for scalability**
- **Limited options for mainstream**
- **Difficulty to transfer capacity & technology to the host nations**



Open-Source Complements

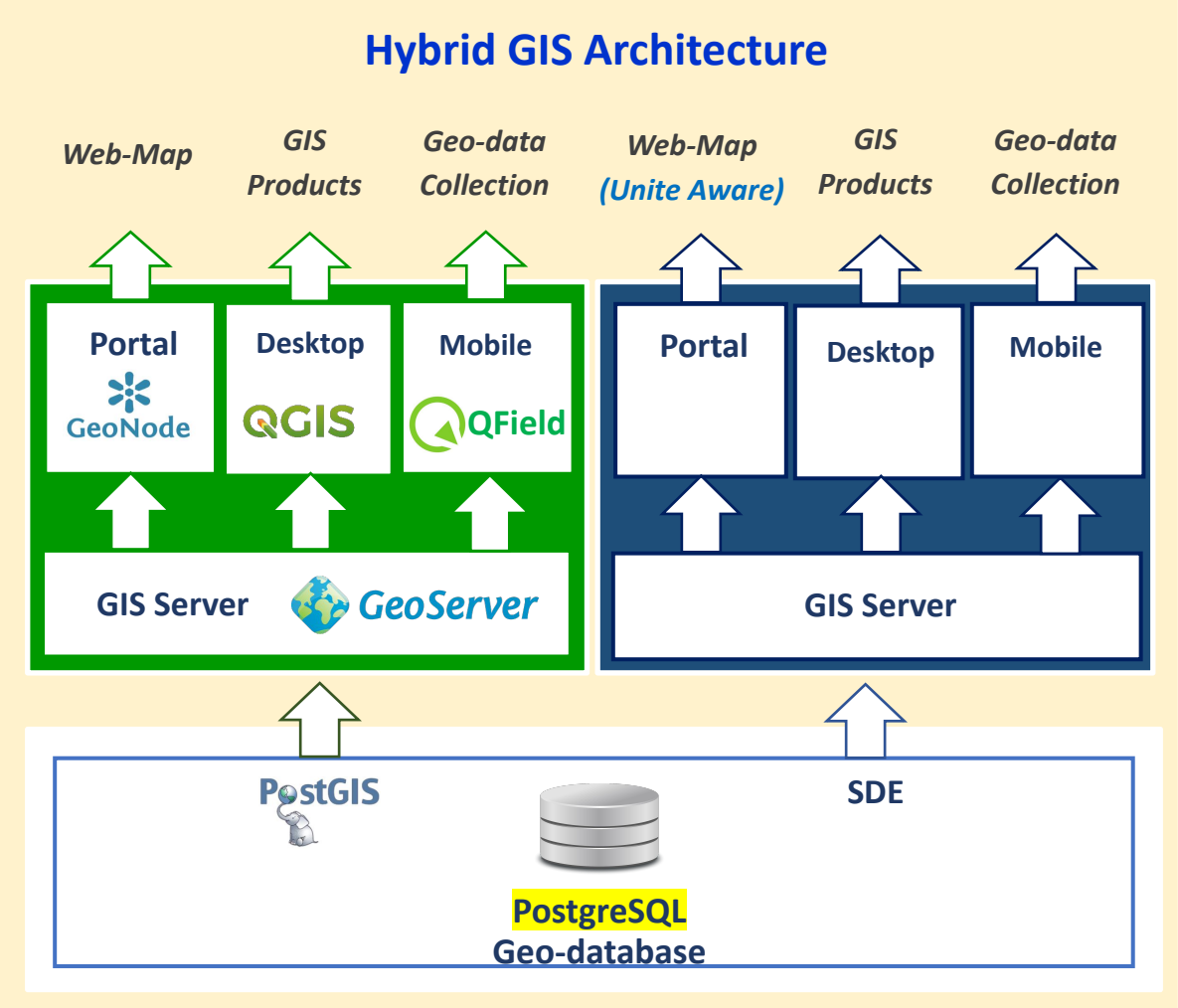
- **OPEN-SOURCE:**
  - Support **operational** and **technical** demands
  - Complementing **proprietary software**
  - **License–free** software
  - **Optimized cost** of running GIS infrastructure
  - **Partially** implemented already (Unite Map)
  - **Flexibility** and **streamlining** of GIS process
  - **Scalability** due to cost efficiency
  - **Interoperability** with the existing infrastructure/ geospatial database
  - **Innovations**
  - **Lighter footprint** on the infrastructure
  - Limitation of pure open–source approach
  - Needed investment to build and maintain Hybrid GIS infrastructure



Hybrid GIS Architecture

Hybrid is the best approach to supporting UN operations

- **HYBRID APROACH:**
  - Integration of Geo-databases to serve proprietary and open-source
  - Complementing systems to serve every UN requirement
  - Cost effective solutions
  - Flexible solutions
  - Scalability
  - Innovations
  - Social value

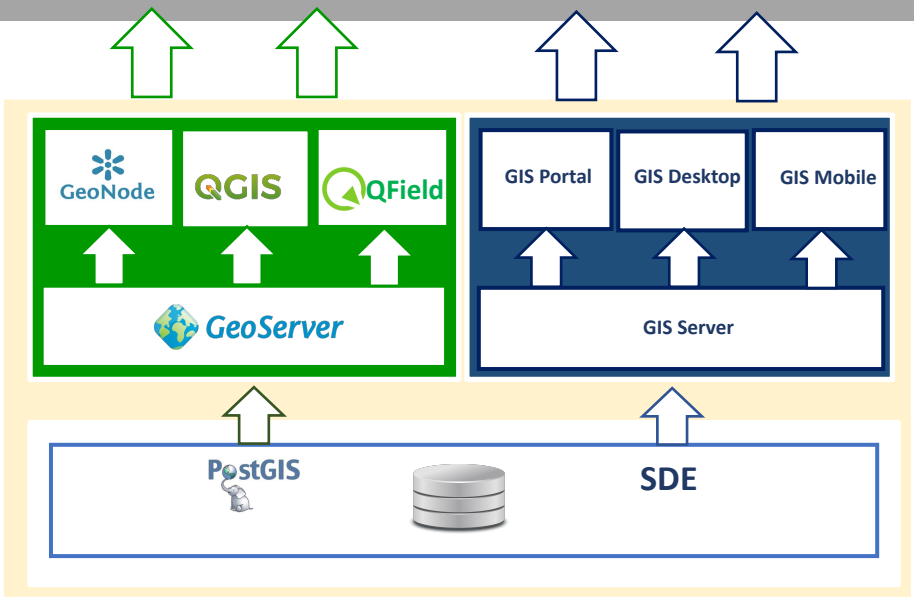


Hybrid GIS Infrastructure

Hybrid GIS will support technology independent UN Geospatial requirements

IMINT	Geo Solutions	Data Services	Geo Products
analysis interpretation SAR IM analysis cloud-free IM IMINT – AI	geo-portal unite aware geo story mobile GIS	unite maps SALB open-drone map UN vector tiles standards policies	analytical prod. thematical prod. web maps printed maps digital maps standards policies

technology independent platforms



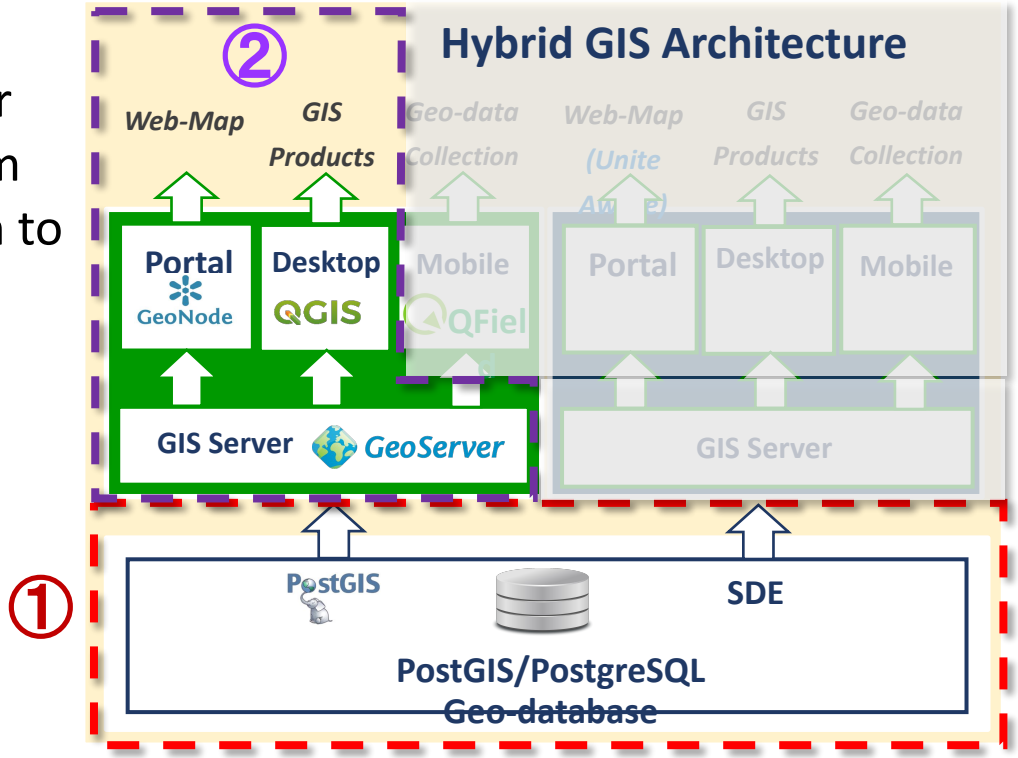
Impact Analysis (Proprietary vs. Hybrid)

	Proprietary system	Hybrid (proprietary and Open-source)
Operation requirements		
Technical requirements		
Finance benefits		
Tech innovations		
Sustainability		
Stability		
Scalability		
Legacy and userbase		
Social value		

Conceptual value (Not a quantifiable measure)

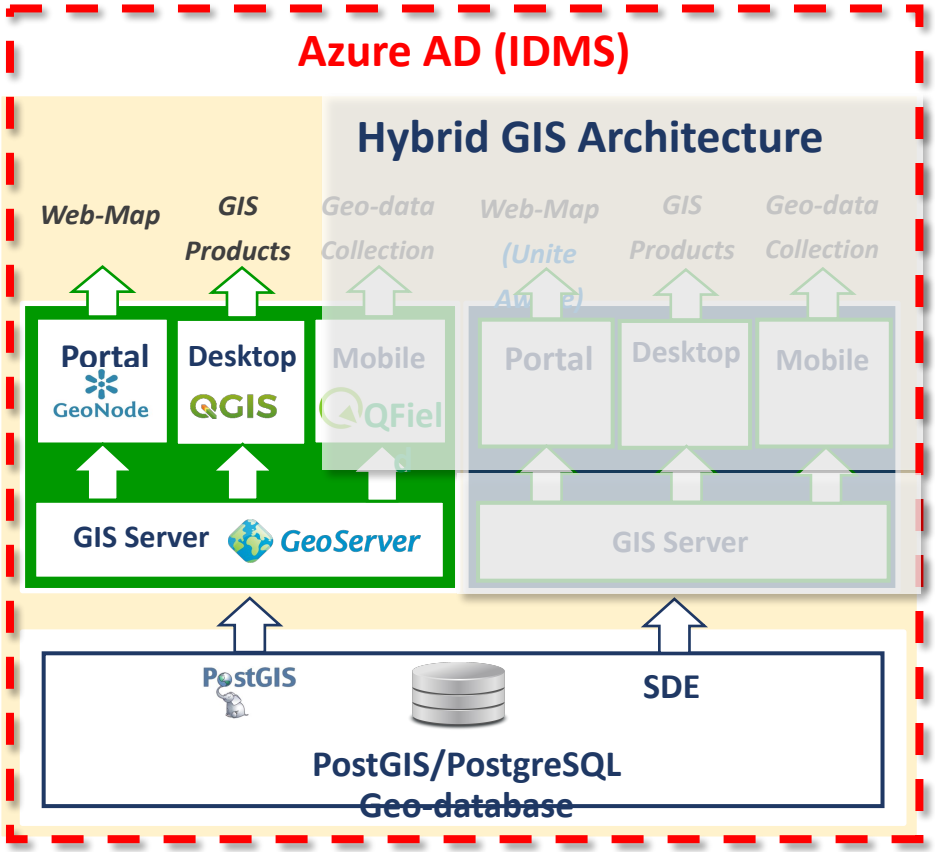
Pilot One Implementation

- **Purpose**: Proof of concept through design & implement ***hybrid prototype*** to support ①**Unite Map**, ②**Open GeoPortal**
- **Contributors**: KRIHS/Korea, UNGIS, UNGSC, WFP
- **Developer**: GeoSolutions
- **Recommendations:**
  Implement UN required User Identity Management System
  Develop a global rollout plan to implement Hybrid GIS infra.



Pilot Two Implementation

- **Purpose**: Integrate Hybrid GIS Infrastructure and User Identity Management System – Azure AD
- **Contributors**: UNGIS, UNGSC
- **Developer**: GeoSolutions
- **Next step (Long-term plan)**:
  To migrate GIS applications to Hybrid GIS



Outcomes:

- **Partial Implementation and operationalization:**

Some components of Hybrid GIS Infrastructure, such as PostgreSQL were implemented. Open GIS desktop software are widely used in UN.

Challenges:

- **Financial:**

Majority of the systems were implemented on Proprietary Platforms. System migration required significant financial recourses.

- **For system development and maintenance**
  - **Infrastructure cost**
  - **To develop in-house capacity**
  - **To outsource development and maintenance**

Questions